MT1011

ENGINEERING MATERIALS

THE COMPLETE MIDTERM STUDY GUIDE

A self-contained, visually guided companion to the **nine supplied lecture PDFs**. Written to help you understand the theory, derive and apply equations, interpret diagrams, and solve examination-style problems without needing to decode the slides.

> **HOW TO USE THIS GUIDE**
> **1.** Read the explanation. **2.** Study the visual/definition. **3.** Follow the method. **4.** Attempt the worked example again without seeing its steps. **5.** Use the end-of-guide checklist and practice questions.

Lecture scope: Introduction I & II; Atomic Bonding; Crystal Structures; Crystal Defects; Mechanical Properties & Measurement; Strengthening Mechanisms; Phase Diagrams; Eutectic Systems.

Priority: Fundamental concepts and calculations receive the most space. The two introduction lectures have been deliberately condensed.

Prepared specifically from the uploaded SLIIT MT1011 lecture set, dated 23 September–27 October 2026. Diagrams are explanatory redrawings (not reproductions of original lecture figures). Values and conventions are identified where they matter.

Your route through the module

Chapter	What you must understand	Lecture slides
01 Materials at a glance	The structure → properties → processing → performance idea; material families	Intro I (10), Intro II (19)
02 Atomic bonding	Energy curves; ionic, covalent, metallic, secondary and mixed bonding	Bonding (31)
03 Crystals and geometry	Seven systems, Bravais lattices, SC/BCC/FCC/HCP, APF, a and r	Crystals (1–40)
04 Crystal directions and planes	Position, Miller indices, close packing; linear and planar densities	Crystals (41–66)
05 Crystal defects	Vacancies, interstitials, dislocations, Burgers vector, slip	Defects (30)
06 Mechanical properties	Tensile curve; yield, true/engineering quantities, ductility, toughness, tests	Mechanical (37)
07 Strengthening	Hall–Petch, solid solutions, strain hardening, percent cold work	Strengthening (28)
08 Isomorphous phase diagrams	Phases, tie lines, lever rule, Cu–Ni cooling and properties	Phase Diagrams (37)
09 Eutectic phase diagrams	Ag–Cu, Pb–Sn, eutectic reaction, microstructures and phase fractions	Eutectics (15)
10 Final-exam toolkit	Question selectors, mistakes, practice and answer checks	All nine lectures

THE MAIN CONNECTION

Bonding controls atomic interactions; **crystal structure and defects** determine movement of atoms/dislocations; **mechanical testing** measures their behaviour; **strengthening** controls defects; **phase diagrams** predict phases and microstructures.

Symbols and units you will meet

Symbol / unit	Meaning	Use
r, a, d	Atomic radius; cell edge length; mean grain diameter	Often nm, mm or m
F, A, l	Applied load; cross-section; gauge length	N, mm^2/m^2, mm/m
σ, ε, E	Stress, strain, elastic modulus	Pa or MPa; no unit; Pa
C_0, C_α, C_L	Overall alloy composition, phase compositions	wt% or at% (use given basis)
W_α, W_L	Mass fractions of phases	Fractions sum to 1
K, °C	Absolute temperature, Celsius	K = °C + 273.15

Symbol / unit	Meaning	Use
1 MPa	1 $N/mm^2 = 10^6$ Pa	Useful in tensile calculations
1 nm	10^{-9} m; 1 m = 10^9 nm	Useful for atomic densities

READ THIS BEFORE STARTING
The study guide follows the supplied lecture level. A few slide phrases/labels are imprecise; where this affects a calculation, the geometrically or physically consistent interpretation is explicitly flagged rather than silently copied.

01

Materials at a glance

A short orientation; deliberately not the main part • Lecture basis: Introduction-I and Introduction-II

What is engineering materials science?

Engineering materials science asks: **which material should we choose, and why will it behave that way?** The answer is not simply “steel is strong” or “plastic is light.” An engineer connects composition and bonding to structure, processing, measured properties and service performance.

CORE CHAIN
Composition + bonding → internal structure → properties → material selection and performance. Changing the processing can change grain size, defects and phases, even without changing the overall composition.

The principal material groups

Material group	Typical behaviour / advantages	Limitations / examples
Metals & alloys	Usually crystalline; good electrical/thermal conduction from mobile electrons; usually ductile and useful for load-bearing parts.	Can corrode; relatively dense in many cases. Fe, Al, Cu, Ti; steel and brass are alloys.
Ceramics	Inorganic, often ionic/covalent; hard, wear resistant, high melting temperature, often stable in hot/harsh environments; frequently insulating.	Often brittle and weak in tension compared with compression. Al_2O_3, SiO_2, SiC, Si_3N_4; traditional clay ceramics.
Polymers	Large carbon-based molecular chains; generally low density, easily formed into complex shapes; many are flexible, chemically inert and insulating.	Usually low stiffness/strength and limited service temperature; soften/decompose. PE, PVC, PS, nylon, PC, silicone rubber.
Composites	Two or more constituents combined to obtain a selected set of properties.	Properties depend on constituents and interfaces. Fiber-reinforced composites, concrete, wood.

Material group	Typical behaviour / advantages	Limitations / examples
Advanced materials	Engineered for high-technology uses: electronic, magnetic, sensor, optical, biomedical applications.	Function is more important than any one broad mechanical-property stereotype.

In the source slides, material comparison charts include **density, elastic modulus, tensile strength and fracture toughness**. These are different properties: low density does **not** automatically mean weak; high stiffness does **not** guarantee toughness; strong ceramics can still fracture suddenly.

> **MEMORISE THESE DISTINCTIONS**
> **Strength:** ability to carry stress before yielding or failing. **Stiffness:** resistance to elastic deformation (E).
> **Ductility:** ability to plastically deform before fracture. **Toughness:** energy absorbed to fracture.
> **Hardness:** resistance to local indentation/scratching.

METHOD | Explain a material choice

01. State the **service requirement** (load, temperature, weight, chemical environment, conductivity, cost).

02. Match it to the **relevant property** (strength, toughness, stiffness, thermal resistance, etc.).

03. Compare at least two candidate material classes, naming both an advantage and a limitation.

04. Conclude with the **best compromise**; there is rarely a universally “best” material.

WORKED EXAMPLE | Property choice

Question. A hot, electrically insulating, wear-resistant component is required. Which broad family would you initially investigate?

Step 1. Convert requirements to properties: thermal stability, electrical insulation and hardness.

Step 2. Ceramics often combine these properties, whereas many polymers soften at modest temperatures and most metals conduct electricity.

Step 3. Select an appropriate engineering ceramic as the first candidate, but check brittleness and impact loading before deciding.

> **RESULT**
> The useful explanation is a property-based argument—not merely naming a material.

02

Atomic bonding: why properties differ

From interatomic force to stiffness, melting and conductivity • Lecture basis: Atomic Bonding in Solids (31 slides)

Atoms attract, repel and settle at an equilibrium distance

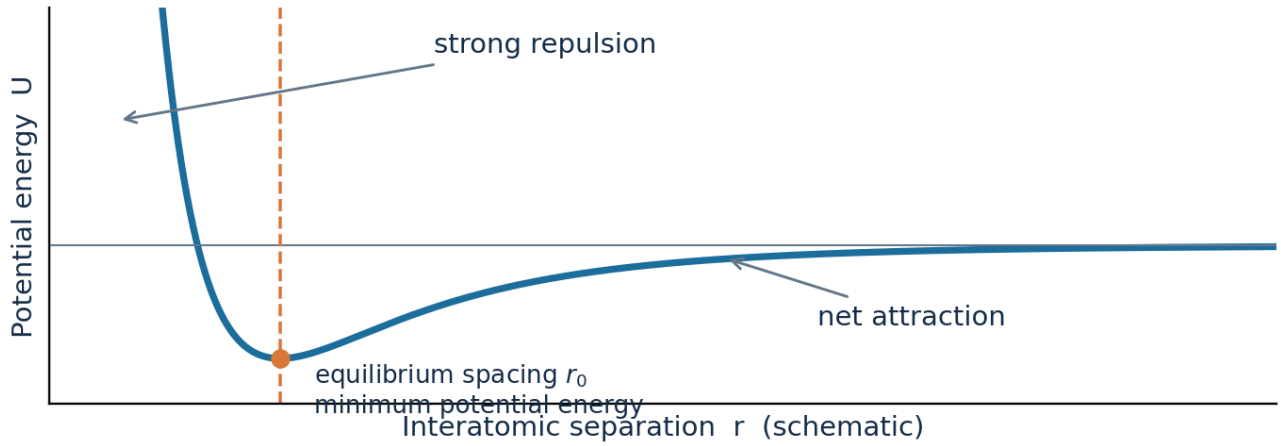
When two atoms are far apart, interaction is weak. As they approach, attractive forces pull them together. When electron clouds strongly overlap at very short separation, repulsive forces rise sharply. The stable spacing is where these effects balance.

$$F_N = F_A + F_R, \qquad F_N(r_0) = 0$$

Symbols: F_N net interatomic force; F_A attractive force; F_R repulsive force (signed); r_0 equilibrium separation.

Condition: The sum uses signed forces: attraction and repulsion have opposite directions.

The equilibrium spacing also corresponds to a **minimum in interatomic potential energy**. The magnitude of the energy required to separate bonded atoms is the **bond energy**. In the lecture, r_0 is of the order of 0.3 nm for many atoms; it is *not* a universal constant.



Schematic potential-energy well. The minimum identifies stable atomic separation; the steep sides explain resistance to stretching/compression.

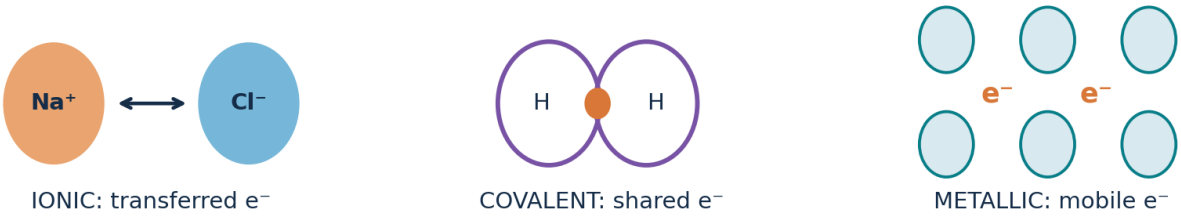
Connect bond energy to familiar properties

- **Melting/boiling:** stronger bonding (larger separation energy) generally requires more thermal energy and gives a higher melting temperature.
- **Elastic modulus E:** a steeper interatomic force–distance curve near equilibrium means greater stiffness; a shallow slope means easy elastic stretching.
- **Thermal expansion:** the asymmetric potential well causes the average separation to grow on heating; a deep, narrow well generally corresponds to lower thermal expansion.
- **Strength ≠ stiffness:** atomic bond stiffness does not directly tell you how readily a real solid yields; dislocations matter later.

VERY COMMON EXAM QUESTION

Why can a material have a high melting point and high stiffness? Explain that large bond energy resists separating atoms, while a steep force–distance slope resists a small change in atomic spacing. Those are related but distinct features of the bond-energy curve.

Primary bonds: ionic, covalent and metallic



Three atomic bonding models: transfer, sharing and delocalization of valence electrons.

Bond	What happens	Direction / properties / examples
Ionic	A metal transfers electron(s) to a nonmetal. Oppositely charged ions attract electrostatically.	**Nondirectional**; generally hard, brittle and insulating in solid state; high melting temperature. NaCl, many ceramics.
Covalent	Nearby atoms share electron pair(s) to achieve stable electron configurations.	**Directional**. Often insulating or semiconducting; mechanical behaviour varies. H_2, Cl_2, CH_4; diamond, Si, Ge, GaAs, SiC.
Metallic	Positively charged ion cores are held together by a mobile, delocalized valence-electron sea.	**Nondirectional**; high electrical/thermal conductivity and generally ductile room-temperature behaviour. Cu, Fe, Al, most alloys.

Ionic sequence (NaCl): Na loses its outer electron and becomes Na^+; Cl gains it and becomes Cl^-. The opposite charges attract. The lecture associates ionic energies with roughly **600–1500 kJ/mol** and high melting points. Ionic compounds can conduct when molten or in solution (because ions can move), even though many are insulating as solids.

Covalent sequence (H_2): each H contributes one electron to a shared bonding pair. Shared electrons sit between particular atoms, hence directional bonding. Diamond is exceptionally hard while silicon and germanium are semiconductors; this shows why one bond type does not guarantee one property.

Metallic sequence: valence electrons can move through the entire crystal instead of belonging to a particular atom. Electron motion helps conduct both heat and electricity. A dislocation can rearrange metallic bonds successively, helping explain common metallic ductility.

Secondary bonds: van der Waals and hydrogen bonding

Secondary bonding is generally weaker than primary bonding and often acts **between** molecules that already have stronger bonds **within** the molecule. It relies on electrostatic attraction between dipoles (unevenly distributed charges).

Interaction	What creates the attraction	Remember
Instantaneous / induced dipole	Fluctuating electron density briefly polarizes one atom, then induces a nearby dipole.	Occurs even in nonpolar substances and noble gases; generally the weakest type.
Permanent dipole → induced dipole	A polar molecule distorts the electron distribution of a nearby nonpolar one.	Typically stronger than purely fluctuating dipole interactions.
Permanent dipole → permanent dipole	Positive end of one polar molecule attracts negative end of another.	Stronger directional intermolecular electrostatic attraction.
Hydrogen bonding	Especially strong interaction where H is covalently bonded to F, O or N and attracts another electronegative site.	Explains relatively high boiling points of H_2O, HF and NH_3.

DISTINGUISH
Inside H_2O: O–H bonds are covalent. **Between water molecules:** hydrogen bonding is a secondary interaction. They are not the same bond.

Mixed ionic–covalent bonds and percent ionic character

Real chemical bonds are not always 100% ionic or 100% covalent. Electronegativity difference gives an approximate ionic character, as used in the lecture.

$$\%IC = \left[1 - e^{-0.25(X_A - X_B)^2}\right] \times 100\%$$

Symbols: %IC = percent ionic character; X_A, X_B = Pauling electronegativities of the two atoms (take the magnitude of their difference).

METHOD | Calculate percent ionic character

01. Read the two electronegativities X_A, X_B from the supplied data.

02. Calculate their difference $\Delta X = |X_A - X_B|$.

03. Square the difference, multiply by −0.25 and evaluate the exponential.

04. Subtract the exponential from 1; convert to a percentage. Check that the value is between 0% and 100%.

WORKED EXAMPLE | Mixed bonding calculation

Question. Suppose two atoms have electronegativities $X_A = 3.0$ and $X_B = 1.0$. Estimate the percent ionic character.

Step 1. $\Delta X = |3.0 - 1.0| = 2.0$.

Step 2. Exponent $= -0.25(2.0)^2 = -1.0$.

Step 3. $\%IC = (1 - e^{-1}) \times 100 = (1 - 0.3679) \times 100 \approx 63.2\%$.

RESULT
Approximately **63.2% ionic character**. This is a model estimate, not a declaration that 63.2% of atoms become ions.

Fast recognition: what bonding question am I seeing?

Question wording	What to discuss
“Why is metal conductive?”	Electron sea; mobile, delocalized valence electrons.
“Why is a ceramic brittle?”	Direction/charge requirements and difficulty of easy slip; avoid claiming all ceramics identical.
“Why high melting temperature?”	Strong atomic bonding / high bond-separation energy.
“Why lower thermal expansion?”	Deep, narrow interatomic energy well.
“Explain H bonding / water boiling”	Strong secondary interaction between polar molecules containing H bonded to N/O/F.
“Bond type from elements?”	Metals + nonmetals often ionic; nearby nonmetals often covalent; metals metallic; real bonds can be mixed.

Source-lecture numeric anchors worth remembering

Type / example	Figure stated in the slides	What it illustrates
Ionic bonding	Approximately 600–1500 kJ/mol	Large bond energy, often high melting temperature
Metallic bonding: mercury	~62 kJ/mol; melting point ≈ −39°C	Metal bonding can be relatively weak
Metallic bonding: tungsten	~850 kJ/mol; melting point ≈ 3414°C	Metal bonding can also be extremely strong
Secondary bonds	Roughly a few to a few tens of kJ/mol	Typically much weaker than primary atomic bonds
Covalent diamond	High hardness; lecture quotes ~3550°C	Strong directional C–C network (temperature is source-reported)

WHY THE VALUES MATTER
These examples illustrate trends rather than universal rules. **Bond type alone** cannot predict every mechanical property; real crystal structures and defects also matter.

03

Crystal structures and atomic packing

Understand the unit cell rather than memorising isolated numbers • Lecture basis: Crystal Structures, slides 1–40

Crystalline or amorphous?

Crystalline: atoms are arranged in a repeating, ordered three-dimensional pattern (long-range periodicity). **Amorphous / non-crystalline:** no repeating long-range arrangement, as in common glass. Some polymers contain both crystalline regions and non-crystalline regions.

The **lattice** is a periodic array of points. A **unit cell** is a repeatable structural block whose translations fill space. Its geometric description uses edge lengths **a, b, c** and angles **α, β, γ**, called the lattice parameters/constants.

Seven crystal systems versus 14 Bravais lattices

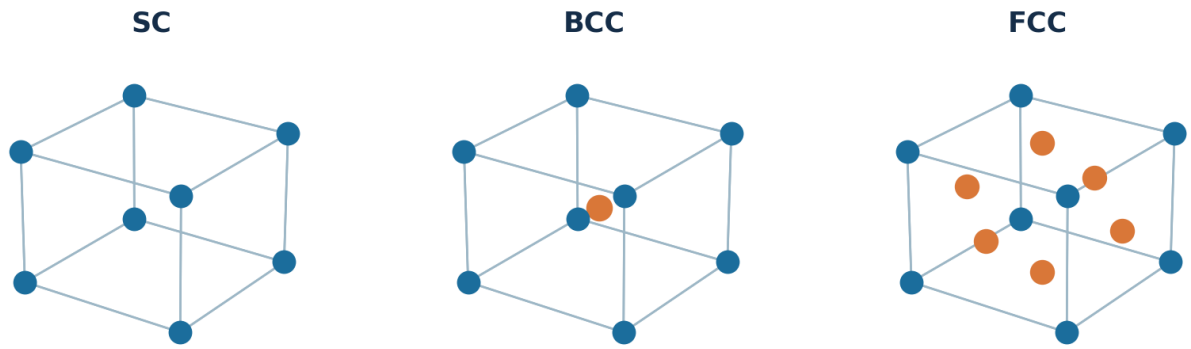
A **crystal system** describes the geometry or symmetry of unit-cell dimensions. A **Bravais lattice** also specifies which lattice-point centerings are allowed. There are seven systems and fourteen Bravais lattices in three dimensions.

System	Edge lengths and angles	Bravais point centerings
Cubic	a = b = c; α = β = γ = 90°	P (primitive), I (body-centered), F (face-centered)
Tetragonal	a = b ≠ c; all 90°	P, I
Orthorhombic	a ≠ b ≠ c; all 90°	P, C (base-centered), I, F
Hexagonal	a = b ≠ c; α = β = 90°, γ = 120°	P
Trigonal / rhombohedral	a = b = c; α = β = γ ≠ 90°	R
Monoclinic	a ≠ b ≠ c; two angles 90°, one ≠ 90°	P, C
Triclinic	a ≠ b ≠ c; angles generally unequal, none required 90°	P

REMEMBER
SC/BCC/FCC are **specific cubic crystal structures**; not all fourteen Bravais lattices are frequently found in elemental metals. HCP uses a hexagonal crystal cell and a multiatom basis.

Hard-sphere model and effective atom counts

To count atoms inside a unit cell, multiply each visible atom by the portion contained within the chosen cell: **corner = 1/8** in cubic cells; **face center = 1/2**; **body center = 1**. For a conventional HCP hexagonal prism, each corner gives 1/6, the two face-center atoms give 1/2 each, and each of three middle-layer atoms contributes 1.



Conventional cubic cells. Blue = corner sites; orange = body- or face-center sites.

Feature	SC	BCC	FCC	HCP (conventional)
Atoms / cell	1	2	4	6
Coordination number	6	8	12	12
Touching direction	Cube edge	Body diagonal	Face diagonal	Close-packed basal direction
a in terms of r	2r	4r / √3	2√2r	a = 2r (basal)
APF	0.524	0.680	0.740	0.740 (ideal)
Stacking	Simple layers	Less closely packed	ABCABC…	ABAB…
Examples	Polonium	α-Fe, Cr, W, Mo	Al, Cu, Ni, Ag, Au	Mg, Ti (ambient), Zn

Coordination number counts the nearest-neighbour atoms touching a reference atom. **Atomic packing factor (APF)** is the fraction of the geometric cell volume occupied by ideal hard spheres.

$$\mathrm{APF} = \frac{n\left(\frac{4}{3}\pi r^3\right)}{V_{\mathrm{cell}}}$$

Symbols: n = effective atoms within unit cell; r = atomic radius; V_{cell} = geometric cell volume (a^3 for cubic).

Condition: The hard-sphere model assumes non-overlapping spherical atoms.

Geometry you should be able to derive

SC: atoms touch along the edge. One edge is two radii, so a = 2r. **BCC:** the body diagonal has length √3a and passes through corner → center → corner, spanning 4r. Hence √3a = 4r. **FCC:** the face diagonal has length √2a and spans 4r; hence √2a = 4r.

$$\mathrm{SC}:\ a = 2r \qquad \mathrm{BCC}:\ \sqrt{3}\,a = 4r \qquad \mathrm{FCC}:\ \sqrt{2}\,a = 4r$$

Symbols: a = cubic unit-cell edge; r = atomic radius; face diagonal = √2a; body diagonal = √3a.

SLIDE TYPO ALERT
One FCC slide mentions a “body diagonal” in its sentence, but the illustrated Pythagoras relation is $a^2 + a^2 = (4r)^2$. **FCC atoms touch along the face diagonal**. BCC atoms touch along the body diagonal.

METHOD | Solve an atomic packing / unit-cell question

01. Identify SC, BCC, FCC or HCP from the atom positions or the given name.

02. Count effective atoms n using shared corner/face/body contributions.

03. Find the touching line to obtain the relationship between a and r.

04. Calculate $V_{atoms} = n(4/3)\pi r^3$ and $V_{cell} = a^3$ for cubic structures.

05. Calculate APF = V_{atoms}/V_{cell}; cancel r^3 before rounding. Interpret as percentage occupied.

WORKED EXAMPLE | BCC lattice parameter and APF

Question. A BCC metal has radius r = 0.124 nm. Find its unit-cell edge a and APF.

Step 1. BCC has two effective atoms and its body diagonal touches four atomic radii.

Step 2. $a = 4r/\sqrt{3} = 4(0.124)/1.732 \approx 0.2864$ nm.

Step 3. $APF = [2(4\pi r^3/3)]/a^3$. Substituting $a = 4r/\sqrt{3}$ gives $(\sqrt{3}\pi)/8 = 0.680$.

RESULT
Unit-cell edge ≈ **0.286 nm**; about **68.0%** of the ideal cubic cell is occupied by atoms.

Layer stacking: the conceptual difference between FCC and HCP

On a close-packed atomic layer, atoms form triangular arrangements. The next layer rests in gaps, giving A and B positions. A third layer can align directly over A or use a new C arrangement. **ABAB… gives HCP**; **ABCABC… gives FCC**. Both reach APF ≈ 0.74 and coordination number 12. The difference is the stacking *sequence*, not the nearest-neighbour count.

For the conventional HCP cell the lecture counts **12 corner × 1/6 = 2**, **2 face centers × 1/2 = 1** and **3 internal atoms × 1 = 3**, totalling **6 atoms**.

WORKED EXAMPLE | A simple atom-count check

Question. How many atoms belong to one conventional FCC unit cell, and why?

Step 1. There are eight corner atoms shared among eight cells: 8 × 1/8 = 1.

Step 2. There are six face-center atoms shared between two cells: 6 × 1/2 = 3.

Step 3. Total = 1 + 3 = 4 effective atoms in the conventional cell.

RESULT
Count **effective contributions**, not every whole sphere drawn.

04

Crystallographic directions, planes and densities

The notation is simple once the coordinate operation is automatic • Lecture basis: Crystal Structures, slides 41–66

Lattice positions: write where the point is

A point inside a crystal unit cell is located in a **right-handed coordinate system**, taking distances as fractions or multiples of the lattice lengths a, b, c. For example, (1/2, 1/2, 0) is halfway along x and y at the bottom face. Commas denote coordinates; do not confuse positions with plane indices.

Crystal directions [uvw]: subtract head minus tail

A **direction** is a vector. Its indices [u v w] are the smallest integer ratio of x, y and z components, taken in units of lattice parameters. A negative index is displayed with a bar, e.g. [$1\bar{2}$ 0]. Equivalent symmetry-related directions form a family, written ⟨u v w⟩.

METHOD | Find direction indices from endpoints

01. Locate two points on the directed line: head (x_2, y_2, z_2) and tail (x_1, y_1, z_1).

02. Subtract **head − tail** separately: ($x_2 - x_1$, $y_2 - y_1$, $z_2 - z_1$).

03. Multiply all three values by a common factor to **clear fractions**; then reduce to the smallest whole-number ratio.

04. Enclose in **square brackets [uvw]**; place a bar above any negative index. A reverse arrow reverses all signs.

WORKED EXAMPLE | Direction indices from an arbitrary line

Question. A line travels from (1/2, 1, 0) to (0, 0, 1). Determine its direction.

Step 1. Head − tail = (0,0,1) − (1/2,1,0) = (−1/2, −1, 1).

Step 2. Multiply each component by 2 → (−1, −2, 2).

Step 3. Use square brackets, with negative signs or bars: [−1 −2 2].

RESULT
[−1 −2 2] (equivalently with overbars on 1 and 2).

DIRECTION RULES
[100] and [200] represent the same **direction** after reducing their ratio; [100] and [−100] point in **opposite** directions. Angle brackets ⟨110⟩ represent a family of symmetry-equivalent directions.

Miller plane indices (hkl): reciprocal of intercepts

A plane has geometric intercepts with the coordinate axes. Miller indices describe its orientation by reciprocating those intercepts. An axis that the plane never meets has infinite intercept; reciprocal = 0. A plane through the origin needs a parallel translation of the coordinate origin before intercepts are read.

METHOD | Find the Miller indices of a plane

01. Identify the intercepts with x, y, z in units of a, b, c (shift the origin to another unit cell if the plane passes through it).

02. Take the reciprocal of **each** intercept: h = 1/x, k = 1/y, l = 1/z. A parallel axis gives ∞ → 0.

03. Multiply by the common denominator to clear fractions (follow your lecture convention; do not reduce indexed planes indiscriminately).

04. Write **(hkl)** in parentheses. Negative intercept → overbar on the corresponding index. Curly braces **{hkl}** mean an equivalent family of planes.

WORKED EXAMPLE | Miller indices with a parallel axis

Question. A plane intercepts x at a, y at 2b, and is parallel to z. Find its Miller indices.

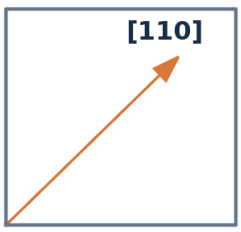
Step 1. Normalize intercepts to lattice units: (1, 2, ∞).

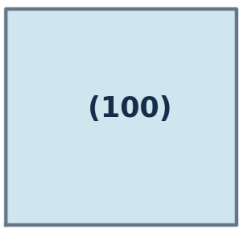
Step 2. Reciprocals: (1, 1/2, 0).

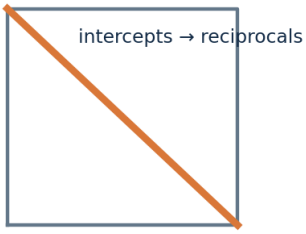
Step 3. Clear the fraction by multiplying all entries by 2 → (2,1,0).

Step 4. Put the indices in round brackets → (210).

> **RESULT**
> **(210)**; the zero shows the plane is parallel to the z axis.







Square brackets describe an arrow (direction); round brackets describe plane orientation. The drawn squares are simplified 2D guides, not complete 3D planes.

Notation	Meaning	Important distinction
(x, y, z)	Coordinates of a lattice **position**	Numbers are actual fractional locations.
[u v w]	One directed vector	Opposite sign → opposite direction; multiples give same direction.
⟨u v w⟩	Symmetry-equivalent **direction family**	Cubic symmetry relates [100], [010], [001].

Notation	Meaning	Important distinction
(h k l)	One **crystallographic plane orientation**	Derived from reciprocals of intercepts.
{h k l}	Symmetry-equivalent **plane family**	Use curly braces only for families.

IMPORTANT LECTURE CONVENTION
The slides emphasize that direction indices are reduced to integer ratios, while plane indices are reciprocals of intercepts and should **not** be casually reduced as if they were direction vectors. Opposite parallel planes are treated as parallel orientations in the lecture; avoid confusing that with opposite *directions*.

Linear density: atoms centered along a chosen direction

The lecture defines **linear atomic density** as the number of atom centers lying on a chosen crystallographic line divided by the length of the line segment. Endpoints count fractionally if they are shared with neighbouring repeated segments. The densest direction is the close-packed direction.

$$\rho_L = \frac{N_{\text{line}}}{L_{\text{segment}}}$$

Symbols: ρ_L = linear density (atoms/nm or atoms/m); N_{line} = effective number of centered atoms on the line; L_{segment} = line length.

Condition: Only count atom centers lying ON the direction, not nearby spheres.

METHOD | Find a linear density in a cubic cell

01. Choose a line segment parallel to the required [uvw] direction that passes through relevant atom centers.

02. Count atom centers along it, including endpoint fractions (typically 1/2 at each repeated end).

03. Calculate segment length: along [100] = a, [110] = √2a, [111] = √3a for a cubic unit cell.

04. Divide effective count by length; compare candidate directions using the same units.

WORKED EXAMPLE | FCC Ni linear density

Question. For FCC Ni with a = 0.35167 nm, calculate linear density along [110].

Step 1. A face diagonal passes through 1 face-center atom, plus 2 shared corner endpoints.

Step 2. Effective number N = 1/2 + 1 + 1/2 = 2 atoms.

Step 3. Segment length L = √2 a = 1.4142 × 0.35167 nm ≈ 0.49734 nm.

Step 4. ρ_L = 2 / 0.49734 ≈ 4.021 atoms/nm.

RESULT
4.021 atoms/nm. FCC ⟨110⟩ directions are close packed.

FCC direction	Effective atom count	Segment length	Linear density
[100]	1	a	1/a
[110]	2	√2a	√2/a ← maximum
[111]	1	√3a	1/(√3a)

Planar density: atoms centered on a chosen plane

Planar atomic density is atom centers lying in a crystallographic plane divided by the plane area. Only count atoms whose centers lie in that plane. Atoms on an edge or corner of the chosen planar repeat region contribute fractional shares.

$$\rho_P = \frac{N_{\text{plane}}}{A_{\text{plane}}}$$

Symbols: ρ_P = planar density (atoms/nm² etc.); N_{plane} = effective number of atom centers on plane; A_{plane} = area of selected planar region.

METHOD | Find FCC planar density

01. Sketch the planar cut for the required plane (100), (110) or (111).

02. Choose a simple planar repeating region (square, rectangle or triangle).

03. Count every atom with its center in that region using appropriate shares of corners/edges.

04. Find the region area from cell geometry; divide atom count by area and identify the densest plane.

FCC plane	Planar geometry from lecture	Atoms N	Planar density
(100)	Square area a²	2	2/a²
(110)	Rectangle area √2a²	2	√2/a²
(111)	Triangle area (√3/2)a²	2	4/(√3a²) ← maximum

WORKED EXAMPLE | FCC Ni close-packed plane

Question. For FCC Ni with a = 0.35167 nm, determine planar density on (111).

Step 1. The lecture uses a triangular (111) region: 3 vertices each contribute 1/6 and 3 edge atoms each contribute 1/2; N = 2.

Step 2. Its area is A = (√3/2)a² ≈ 0.8660(0.35167)² ≈ 0.10711 nm².

Step 3. ρ_P = 2/A = 4/(√3 a²) ≈ 18.67 atoms/nm².

> **RESULT**
> **18.67 atoms/nm².** The closest-packed FCC plane is {111}.

CHECKPOINT
Know the FCC close-packed **direction ⟨110⟩** and **plane {111}**. The atom-count and geometry procedure is more reliable than trying to recall an isolated answer.

05

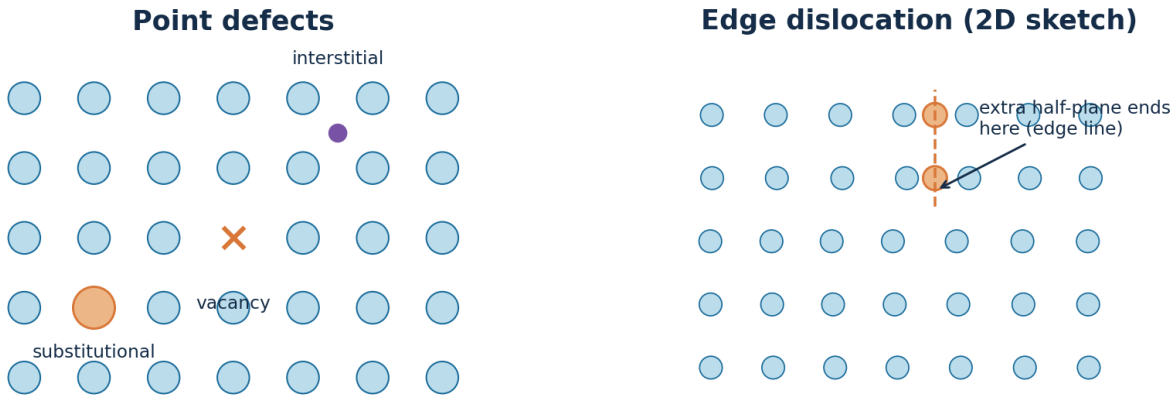
Crystal defects, dislocations and slip

Imperfections are the reason real metals deform • Lecture basis: Crystal Defects (30 slides)

Four scales of imperfection

A perfect crystal is an idealization. Real solids contain imperfections that can influence strength, deformation, diffusion and surface behaviour. The classification comes from defect *dimensionality*.

Defect class	Dimension	Main examples
Point defects	Localized in all three directions	Vacancy, substitutional solute, interstitial solute, self-interstitial
Line defects	Extended along a line	Edge, screw and mixed dislocations
Planar / surface defects	Extended over a surface	Grain boundaries, phase boundaries, external surfaces
Bulk defects (context only)	Three-dimensional region	Porosity, inclusions; not developed in the uploaded defect lecture



Point-defect lattice and a schematic extra half-plane ending at an edge-dislocation line.

Point defects: four important names

- **Vacancy:** a missing atom at a normally occupied lattice site. Its equilibrium concentration rises strongly as temperature increases.
- **Substitutional impurity:** a foreign/solute atom replaces a host/solvent atom at a normal site; common for similarly sized metal atoms.
- **Interstitial impurity:** a small foreign atom sits in a space between regular sites; C, N and O are important examples in metals.

- **Self-interstitial:** an extra host atom occupies a normally empty interstitial position; causes a large lattice distortion and is less common than vacancies in metals.

Vacancy concentration rises exponentially with temperature

$$N_V = N\exp\left(-\frac{Q_v}{kT}\right)$$

Symbols: N_v = number of equilibrium vacancies; N = number of available atomic sites; Q_v = energy per vacancy; k = Boltzmann constant; T = absolute temperature in kelvin.

Condition: Use compatible energy units: k = 8.62 × 10^{-5} eV/(atom·K) when Qv is in eV/atom, or 1.38 × 10^{-23} J/(atom·K) when Qv is in joules/atom.

The vacancy fraction is $N_v/N = \exp[-Q_v/(kT)]$. The lecture notes approximately **10^{-4}** near the melting temperature in many metals—about one missing site per 10,000 sites. Increasing T makes the negative exponent less negative, so the vacancy population increases.

$$N = \frac{\rho N_A}{M}$$

Symbols: N = atomic sites per unit volume (atoms/m^3 or atoms/cm^3); ρ = material mass density; N_A = Avogadro constant ≈ 6.022 × 10^{23} mol^{-1}; M = molar mass.

Condition: Use ρ and M with consistent length/mass units. This conversion is needed in the lecture's copper vacancy exercise.

METHOD | Calculate equilibrium vacancies per cubic metre

01. Convert temperature T = θ(°C) + 273.15 K.

02. Choose k in the same energy units as Q_v (eV with eV; joules with joules).

03. Find site density $N = \rho N_A/M$, taking care with $cm^3 \leftrightarrow m^3$ if necessary.

04. Evaluate $f_v = \exp[-Q_v/(kT)]$ and $N_v = N f_v$.

05. Check N_v/N lies between 0 and 1; include the requested volumetric units.

WORKED EXAMPLE | Copper vacancy calculation from the lecture

Question. For Cu at 1000°C, use Q_v = 0.9 eV/atom, M = 63.5 g/mol and ρ = 8.4 g/cm^3. Estimate equilibrium vacancies per m^3.

Step 1. T = 1000 + 273.15 = 1273.15 K.

Step 2. N = (8.4/63.5)(6.022 × 10^{23}) atoms/cm^3 ≈ 7.967 × 10^{22} atoms/cm^3 = 7.967 × 10^{28} atoms/m^3.

Step 3. $f_v = \exp[-0.9/(8.62 \times 10^{-5} \times 1273.15)] \approx 2.74 \times 10^{-4}$.

Step 4. $N_v = (7.967 \times 10^{28})(2.74 \times 10^{-4}) \approx 2.18 \times 10^{25}$ vacancies/m^3.

> **RESULT**
> About **2.18 × 10^{25} vacancies/m^3**. This uses the supplied high-temperature density, not a room-temperature value.

Line defects: edge, screw and mixed dislocations

A **dislocation** is a line of atomic misregistry that can move under a suitable shear stress. Without dislocations, sliding one whole perfect atomic plane at once would require much greater stress. Dislocation motion allows a small region to shift in steps, reducing the observed stress needed for plastic deformation.

Type	Geometric description	Burgers vector relation
Edge	Extra half-plane terminates along a dislocation line.	b is **perpendicular** to the dislocation line.
Screw	Shear distortion resembles a spiral ramp around the line.	b is **parallel** to the dislocation line.
Mixed	Dislocation line locally contains both edge-like and screw-like character.	b has general orientation relative to line.

Burgers vector b: close an atomic-step circuit around a dislocation; the closure failure is the Burgers vector. Its direction and magnitude quantify the lattice displacement associated with the dislocation. In the lecture, an edge dislocation travels **parallel to b**, whereas a screw dislocation line travels **perpendicular to b** during the described motion; distinguish *motion direction* from *line orientation*.

Slip systems: where plastic deformation takes place

A **slip system** is a combination of a **slip plane** and a **slip direction**. Dislocations move most readily on densely packed atomic planes and in highly packed directions. This gives permanent atomic displacement and can leave a step at a free surface.

- **Slip plane:** the crystallographic plane on which dislocation glide takes place.
- **Slip direction:** direction of relative atomic displacement in that plane.
- **Example from the lecture:** aluminum is FCC; its familiar easy slip system combines the {111} family of planes with ⟨110⟩ directions.
- **Why metals are ductile:** many dislocations can move and distribute plastic strain before fracture.
- **Why strengthening is possible:** obstacles to dislocation motion require greater applied stress for the same plastic deformation.

METHOD | Answer a dislocation/slip explanation question

01. Identify defect type and describe its geometry (half-plane, spiral shear, mixed).

02. Name b and its relationship to the dislocation line when relevant.

03. Identify the applied **shear stress** as the driver of dislocation glide.

04. Describe a small atomic rearrangement progressing through the lattice and the resulting permanent slip.

05. Connect easier slip to ductility/lower yield strength, or obstructed slip to higher strength.

Planar defects and boundaries

Grain boundaries separate regions of the same crystal structure having different crystallographic orientations. The atomic arrangement is less regular at the boundary. Such boundaries can obstruct slip and are the reason grain-size refinement often strengthens metals. A **phase boundary** separates distinct

phases. An **external surface** is a truncated lattice: some bonding is incomplete, and the surface can be reactive or contain notches.

> **EXAM DISTINCTIONS**
> A **vacancy** is a missing atom; a **self-interstitial** is an extra host atom squeezed into an interstice. A **grain boundary** separates crystal orientations; a **phase boundary** separates phases. An **edge dislocation** is a line defect—not an entire missing plane.

06

Mechanical properties and how they are measured

The stress–strain graph is a map of material behaviour • Lecture basis: Mechanical Properties & Measurement (37 slides)

A tensile test asks two basic questions

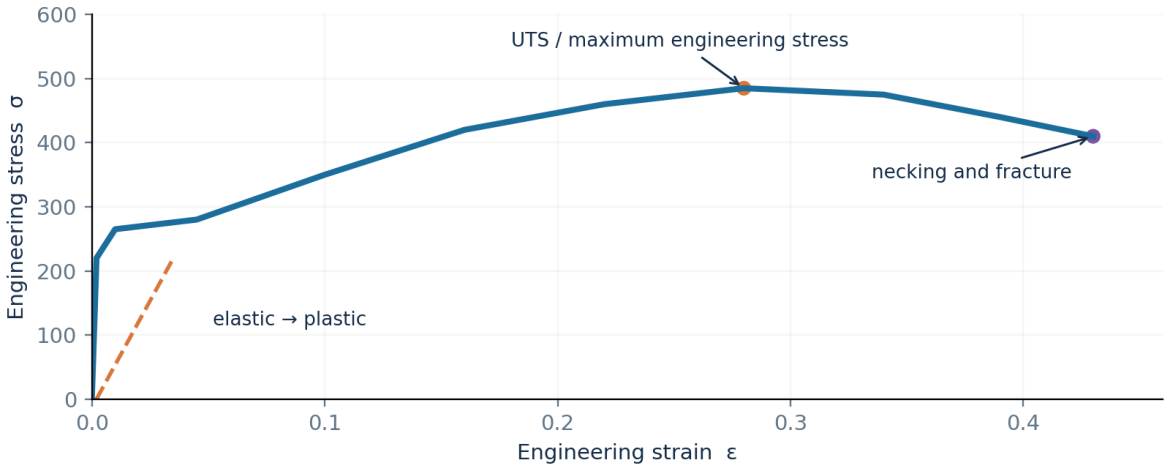
A standardized specimen is held by grips and stretched at a controlled rate. The machine records **load** and **elongation**. Because thick and thin specimens cannot be compared directly from load alone, convert the results to engineering stress and strain using their **original** dimensions.

$$\sigma_e = \frac{F}{A_0} \qquad \varepsilon_e = \frac{l - l_0}{l_0} = \frac{\Delta l}{l_0}$$

Symbols: σ_e = engineering tensile stress; F = applied force; A_0 = initial cross-sectional area; ε_e = engineering strain; l_0 = initial gauge length; l = gauge length during loading; Δl = elongation.

Condition: Stress is often MPa = N/mm²; engineering strain has no unit (multiply by 100 for %).

A positive tensile stress generally causes a positive strain. **Elastic deformation** is fully recoverable upon unloading. **Plastic deformation** leaves permanent strain, although the elastic part still recovers.



Schematic tensile-test curve with elastic loading, transition to plasticity, maximum engineering stress and necking. Exact shapes vary by material.

Find the five key outputs of a tensile test

Property	Where to find / how to calculate	What it means
Young's modulus E	Slope of the initial linear σ–ε segment.	**Stiffness**, resistance to elastic strain.
Yield strength σ_y	Onset of plasticity; often 0.2% offset construction.	Stress for a small specified permanent strain.
Tensile strength TS / UTS	Highest engineering stress = F_{max}/A_0.	Maximum engineering stress in the test.
Ductility	Percent elongation %EL and/or percent reduction in area %RA.	Extent of permanent deformation before fracture.
Toughness	Area under entire stress–strain curve to fracture.	Energy absorbed per unit initial volume to fracture.

Young's modulus: slope in the elastic region

$$\sigma = E\varepsilon \qquad E = \frac{\Delta\sigma}{\Delta\varepsilon}$$

Symbols: σ = uniaxial stress; ε = elastic strain; E = elastic/Young's modulus (same stress units as σ).

Condition: Hooke's law holds in the linear elastic portion; do not use its slope to describe the whole plastic curve.

WORKED EXAMPLE | Engineering stress and elastic extension

Question. A specimen has A_0 = 50 mm², l_0 = 100 mm, load F = 10,000 N and E = 200 GPa. Assume linear elastic loading. Find stress, strain and extension.

Step 1. Engineering stress σ = 10,000/50 = 200 N/mm² = 200 MPa.

Step 2. Convert E = 200 GPa = 200,000 MPa.

Step 3. Strain ε = σ/E = 200/200,000 = 0.001.

Step 4. Elongation $\Delta l = \varepsilon l_0$ = 0.001 × 100 mm = 0.100 mm.

> **RESULT**
> Stress = **200 MPa**, strain = **0.001**, extension = **0.10 mm** (provided the stress remains below yield).

Yield strength: the 0.2% offset / proof-stress method

Many metallic curves turn smoothly from elastic to plastic and have no sharp yield point. To define a reproducible criterion, move along the strain axis to **ε = 0.002 (0.2%)**, draw a line **parallel to the initial elastic line**, and read the stress where that line intersects the measured curve. This is the 0.2% offset yield strength.

METHOD | Construct yield from a stress–strain graph

01. Identify the straight, elastic segment and its slope E.

02. Mark **ε = 0.002** along the horizontal axis (check axis scaling; 0.2% is not 0.2 strain).

03. Through that point draw a line parallel to the initial elastic segment.

04. At its intersection with the measured curve, read the ordinate σ_y. Label the offset construction.

Some low-carbon steels show an **upper yield point** followed by a small drop and a **lower yield plateau**. In that case the lecturer uses the average stress associated with the **lower yield point** as yield strength; a 0.2% offset construction is not necessary when yielding is clearly defined.

Ultimate strength, necking and fracture

After yield, strain hardening usually raises the engineering stress to a maximum. At the maximum engineering stress, **necking** begins: deformation becomes concentrated into a shrinking cross-section. The load may then fall. The curve reaches **fracture** after more localization.

$$\mathrm{TS} = \frac{F_{\max}}{A_0} \qquad \sigma_{f,e} = \frac{F_f}{A_0}$$

Symbols: TS = tensile strength (ultimate engineering stress); F_{max} = maximum measured tensile load; $\sigma_{f,e}$ = engineering fracture stress; F_f = load at fracture; A_0 = original cross-section.

Do not conclude that the metal has become intrinsically easier to deform because engineering stress falls after UTS. Engineering stress still divides by A_0, even though the real neck area is becoming much smaller. **True stress** uses the instantaneous area and can keep rising until fracture.

$$\sigma_{\text{true}} = \frac{F}{A_{\text{actual}}} \qquad \varepsilon_{\text{true}} = \ln \frac{l}{l_0}$$

Symbols: σ_{true} = true stress; A_{actual} = instantaneous minimum/representative cross-section; ε_{true} = logarithmic true strain; l/l_0 length ratio.

Condition: True uniform strain formula is useful before pronounced necking; avoid treating total gauge length as uniform after necking.

> **DISTINGUISH 4 STRESSES**
> **Yield** starts defined plasticity. **UTS** is highest *engineering* stress. **Fracture engineering stress** uses the fracture load and initial area. **True fracture stress** uses actual area at fracture. These numbers generally differ.

Ductility: percent elongation and reduction in area

$$\%EL = \frac{l_f - l_0}{l_0} \times 100\%$$

Symbols: l_0 = original gauge length; l_f = gauge length measured after fracture when pieces are placed together; %EL = percentage elongation at break.

$$\%RA = \frac{A_0 - A_f}{A_0} \times 100\%$$

Symbols: A_0 = original cross-sectional area; A_f = minimum cross-sectional area at fracture; %RA = percentage reduction in area.

Both measure ductility, but they are not interchangeable. The lecture cautions that **%EL depends on gauge length** (the fraction of extension concentrated near the neck changes with the chosen l_0). Standard gauge length should therefore be stated, commonly 50 mm. A relatively brittle material is described in the lecture as showing fracture strain below about **5%**.

WORKED EXAMPLE | Ductility measured two ways

Question. A test piece starts at l_0 = 50 mm and A_0 = 40 mm²; after fracture its rejoined gauge length is 62 mm and minimum area A_f = 25 mm². Find %EL and %RA.

Step 1. %EL = [(62 − 50)/50] × 100 = 24%.

Step 2. %RA = [(40 − 25)/40] × 100 = 37.5%.

Step 3. Interpretation: both indicate appreciable plasticity, but they measure different geometry changes.

> **RESULT**
> **%EL = 24%**; **%RA = 37.5%**. Never write that the results must be equal.

Unloading and permanent strain

A specimen plastically stretched to a point on the curve still contains an elastic strain component. When the load is removed, it unloads approximately along a line with slope E. The intersection of that line with the **zero-stress strain axis** is the permanent/plastic strain. The amount the strain decreases during unloading is the recovered elastic strain.

METHOD | Find plastic strain from unloading

01. Locate the point at which unloading starts (σ_u, ε_{total}).

02. Use the elastic modulus E to obtain recoverable strain $\varepsilon_{elastic} = \sigma_u/E$.

03. Subtract: $\varepsilon_{plastic} = \varepsilon_{total} - \varepsilon_{elastic}$.

04. On a graph, draw the unloading line parallel to the initial elastic slope and identify its ε-axis intercept.

WORKED EXAMPLE | Permanent deformation

Question. At an unloading point, total engineering strain is 0.015, applied stress is 300 MPa and E = 150 GPa. Estimate permanent strain.

Step 1. Convert E = 150,000 MPa.

Step 2. Elastic recovered strain = σ/E = 300/150,000 = 0.002.

Step 3. Permanent strain = 0.015 − 0.002 = 0.013.

RESULT
Plastic strain = **0.013 = 1.3%** after unloading, within the usual linear unloading approximation.

Resilience, toughness and brittle behaviour

Modulus of resilience U_r is elastic energy stored per unit volume up to yielding: the *area under the elastic region*. For a linear elastic segment it forms a triangle. **Toughness** is the energy per unit volume absorbed up to fracture: the *total* stress–strain area. These are not the same.

$$U_r = \frac{1}{2}\sigma_y\varepsilon_y = \frac{\sigma_y^2}{2E}$$

Symbols: U_r = modulus of resilience (J/m³ if stress in Pa); σ_y = yield stress; ε_y = strain at yield; E = Young's modulus.

Condition: Linear elastic behaviour up to the chosen yield point. This is a useful geometric consequence of the lecture stress–strain area discussion.

For any irregular curve, toughness is the integral of engineering stress with respect to engineering strain up to fracture. In a graphical question, approximate by summing polygon/trapezoid areas if points are supplied. **High strength by itself is not high toughness:** large ductility contributes greatly to the area.

Ceramics and glasses often show essentially elastic behaviour followed by sudden brittle fracture. They generally sustain much higher compressive stresses than tensile stresses. They may be tested in **bending** rather than direct tension, where the maximum tensile stress on the outer surface triggers failure. **Modulus of rupture (MOR)** is flexural strength, not Young's modulus.

Impact tests: Charpy versus Izod

The **Charpy impact test** swings a weighted pendulum from height h onto a notched specimen. After fracture, it rises to a smaller height h′. The lost potential energy represents impact energy absorbed by the test system/specimen (ignoring test losses). The notch causes high local stress concentration.

$$E_{\text{impact}} \approx mg(h - h')$$

Symbols: m = pendulum mass; g = gravitational acceleration (≈ 9.81 m/s²); h = release height; h′ = rebound height; E_{impact} = absorbed impact energy (J).

Condition: Ideal energy balance; practical machines report the calibrated energy loss, with friction/air losses accounted for.

The source notes the key Charpy/Izod difference is **how the specimen is mounted**. Charpy generally supports a horizontal beam struck centrally; Izod generally clamps it as a vertical cantilever. Both measure impact energy, not the same quantity as a standardized fracture-mechanics toughness K_{IC}.

WORKED EXAMPLE | Pendulum impact energy

Question. A 15 kg pendulum falls from h = 1.10 m and reaches h′ = 0.72 m after breaking the specimen. Estimate energy absorbed ignoring friction.

Step 1. Height loss = 1.10 − 0.72 = 0.38 m.

Step 2. $\Delta PE = mg\Delta h = 15 \times 9.81 \times 0.38 \approx 55.9$ J.

Step 3. Energy is positive because the pendulum has less potential energy after fracture.

> **RESULT**
> Approximate impact energy = **55.9 J**.

Hardness tests and why engineers use them

Hardness is resistance to localized plastic deformation, such as indentation or scratching. An indenter is pressed into a surface using a controlled load/rate; depth or impression size is converted into a hardness value. A deeper/larger imprint usually means softer material.

- **Advantages:** rapid, inexpensive, little specimen preparation, and nearly nondestructive apart from the small indent.
- **Interpretation:** hardness often correlates empirically with tensile strength for a given material family, but there is no universal conversion.
- **Comparing tests:** hardness numbers from different methods are not directly comparable without an appropriate scale/conversion.
- **Common names shown in the lecture charts:** Brinell, Rockwell, and Vickers (identify indenters/load measurement from the specific diagram provided).

> **STUDY THIS CURVE FROM MEMORY**
> For any tensile curve, label: **elastic region → proportional slope E → yield/proof point → plastic strain hardening → UTS → necking → fracture**; mark an unloading path and differentiate engineering versus true stress.

07

How metals are strengthened

All three lecture methods make dislocation motion harder • Lecture basis: Strengthening Mechanisms (28 slides)

Start with one fundamental idea

Most plastic deformation in metallic crystals occurs by **dislocation motion**. If dislocations move easily, a metal is relatively soft and weak; if motion is obstructed, a larger applied stress is required for the same plastic deformation. The lecture develops three major strengthening routes: **grain refinement, solid-solution strengthening and strain hardening**.

> **KEY TRADE-OFF**
> Strength and hardness commonly increase when dislocation movement becomes harder, but **ductility generally falls**. Designers try to achieve a useful balance of strength, toughness and formability.

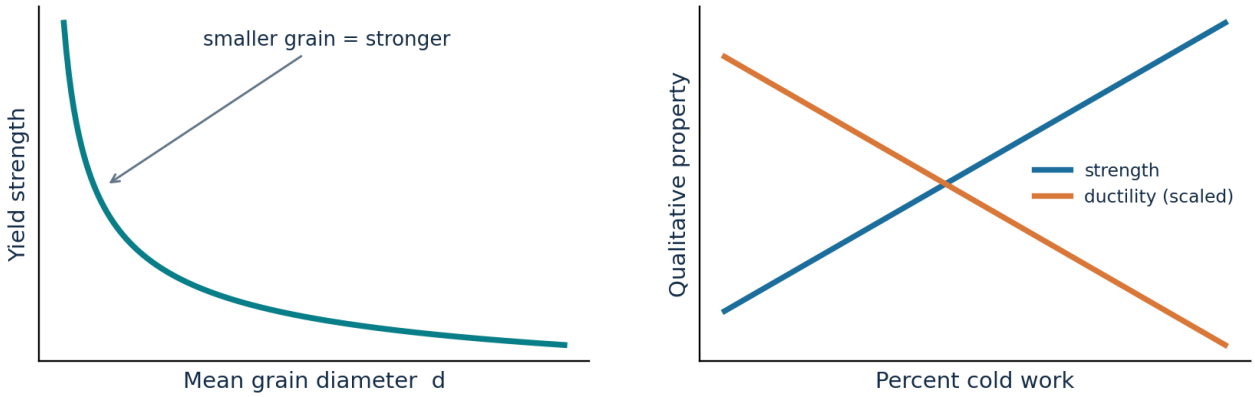
1. Grain size reduction and the Hall–Petch relation

A polycrystalline specimen contains many grains, each with a possibly different crystallographic orientation. A moving dislocation encounters a grain boundary where slip-plane alignment and atomic order are interrupted. Smaller grains mean **more boundary area per volume**, and hence more barriers to slip.

$$\sigma_y = \sigma_0 + \frac{k}{\sqrt{d}}$$

Symbols: σ_y = yield strength; σ_0 = material-dependent intercept/friction stress; k = Hall–Petch strengthening constant; d = average grain diameter.

Condition: As stated in the lecture, the empirical relation does not hold universally at very coarse or extremely fine grain sizes.



Qualitative trends: grain refinement raises yield strength; cold work usually raises strength but lowers ductility. These are schematic, not measured data.

METHOD | Solve Hall–Petch problems

01. Read σ_0, k and d carefully; check that d uses the length unit expected by k (often $mm^{1/2}$ or $m^{1/2}$ in coefficient units).

02. Compute √d, then k/√d, then add σ_0.

03. For comparisons of the *same alloy*, use the same σ_0 and k.

04. For an unknown grain size, rearrange $d = [k/(\sigma_y - \sigma_0)]^2$.

05. Explain physically that finer grains impede dislocation transmission across more boundaries.

WORKED EXAMPLE | Hall–Petch strength prediction

Question. A metal obeys $\sigma_y = \sigma_0 + k/\sqrt{d}$ with $\sigma_0 = 100$ MPa, $k = 0.50$ MPa·√m and $d = 25$ μm. Find σ_y.

Step 1. Convert grain size to metres because k contains √m: $d = 25 \times 10^{-6}$ m.

Step 2. $\sqrt{d} = \sqrt{(25 \times 10^{-6})} = 0.005$ √m.

Step 3. $k/\sqrt{d} = 0.50/0.005 = 100$ MPa.

Step 4. $\sigma_y = 100 + 100 = 200$ MPa.

> **RESULT**
> **σ_y = 200 MPa**. If d were one-quarter as large, the Hall–Petch strengthening term would double.

Grain size can be changed through **solidification**, **plastic deformation** and **heat treatment**. The lecturer notes that grain refinement can also improve toughness in many alloys, although application-specific behaviour may vary.

2. Solid-solution strengthening: lattice strain from impurities

When solute atoms enter a host metal, they produce local lattice distortion. A **substitutional** solute replaces a host atom; an **interstitial** solute fits between host atoms. Their stress fields interact with dislocations and make their motion energetically unfavourable.

- **Small substitutional solute:** tends to impose a tensile misfit strain in surrounding material. The lecture shows such solutes tending toward regions above the edge-dislocation slip plane where they reduce the local compressive strain energy.
- **Large substitutional solute:** tends to impose compressive misfit strain. The lecture shows solutes collecting beneath the slip plane at tensile strain regions.
- **Energy explanation:** a solute sitting near a dislocation can lower the combined elastic strain energy. Moving the dislocation away raises that energy, so more shear stress is needed.
- **Trend:** increasing solute content often raises strength and hardness while reducing ductility; the Cu–Ni system in the slides illustrates this.

> **COMPARE THE THREE METHODS**
> **Fine grains** → more grain-boundary barriers. **Solute atoms** → dislocation–solute strain-field interactions. **Cold work** → more dislocations interfering with one another. These are mechanisms, not merely processing names.

3. Strain hardening / work hardening / cold working

When a ductile metal is plastically deformed at a temperature low enough that recovery is limited, the density of dislocations increases. Their average spacing falls, and their stress fields impede one another. Continued deformation therefore requires a larger stress. The terms **strain hardening**, **work hardening**, and **cold working** are used together in the lecture.

$$\%CW = \frac{A_0 - A_d}{A_0} \times 100\%$$

Symbols: %CW = percent cold work; A_0 = cross-sectional area before cold deformation; A_d = cross-sectional area after deformation.

Condition: For the stated area-reduction measure, compare the area normal to the deformation direction; do not confuse A_d with tensile-test fracture area A_f.

METHOD | Calculate cold work or final area

01. Identify original area A_0 and deformed area A_d (not the fracture surface area from a different tensile problem).

02. Compute %CW = $[(A_0 - A_d)/A_0] \times 100\%$.

03. If given %CW and A_0, rearrange $A_d = A_0[1 - (\%CW/100)]$.

04. Interpret rising %CW as stronger/harder but less ductile, other conditions held constant.

WORKED EXAMPLE | Percent cold work

Question. A metal wire area decreases from 80 mm² to 52 mm² after cold drawing. Find percent cold work.

Step 1. Area reduction = 80 − 52 = 28 mm².

Step 2. %CW = (28/80) × 100% = 35%.

Step 3. Expected trend: higher yield and tensile strengths, higher hardness and lower elongation than before drawing.

> **RESULT**
> Percent cold work = **35%**.

Why does ductility decline as strengthening increases?

Both solutes and additional dislocations obstruct slip. This increases the force required to start or continue plastic flow. When the available mechanisms for redistributing strain are more restricted, the material may accommodate less plastic deformation before fracture. **Annealing** can remove much of the effect of cold work by rearranging and reducing dislocations; the lecture mentions that strain-hardening effects can be removed by suitable heat treatment.

Mechanism	What you change	Why it strengthens	Common trade-off
Grain refinement	Smaller average grain size	More misoriented boundaries interrupt slip	Often ↑ strength, can ↑ toughness

Mechanism	What you change	Why it strengthens	Common trade-off
Solid solution	More/different solute atoms	Solute elastic strain fields pin/interact with dislocations	↑ strength, ↓ ductility
Strain hardening	Greater plastic deformation / %CW	Dislocations multiply and obstruct one another	↑ yield, UTS, hardness; ↓ ductility

METHOD | Answer “which sample is stronger?” diagram questions

01. If micrographs have **the same magnification and alloy**, compare average grain diameter. Finer grains generally imply greater yield strength by Hall–Petch.

02. If the figures compare different impurity contents, recognize greater solute strengthening when the same solid solution persists.

03. If the figures compare cold-work levels, recognize more work hardening with greater area reduction.

04. State the reason using **dislocation mobility**, rather than merely memorizing a trend.

08

Phase diagrams and Cu–Ni isomorphous alloys

Identify phases, read compositions, then calculate quantities • Lecture basis: Phase Diagrams (37 slides)

Terminology you must be able to define

Term	Meaning in the lectures
Component	Pure metal or compound from which an alloy system is made; brass has Cu and Zn components.
System	Set of possible alloys composed of the same components, for example the Cu–Ni binary system.
Phase	Physically and chemically uniform region with distinguishable properties. A solid solution can be a phase.
Solubility limit	Maximum solute concentration that dissolves in the solvent to form one solution at stated conditions.
Homogeneous	One-phase system (in the lecture sense).
Heterogeneous	System with two or more phases.
Microstructure	Spatial arrangement and proportions of phases/grains observed, for example under optical/electron microscope.
Solid solution	One solid phase with dissolved foreign atoms on substitutional or interstitial lattice sites.

Term	Meaning in the lectures
Liquidus	Boundary above which the alloy is completely liquid (in the simple diagrams studied).
Solidus	Boundary below which the alloy is completely solid.
Solvus	Solid-state solubility boundary between single solid-solution and two-solid-phase regions.
Tie line	Horizontal constant-temperature line across a two-phase region; its endpoints show equilibrium phase compositions.

What does a phase diagram actually answer?

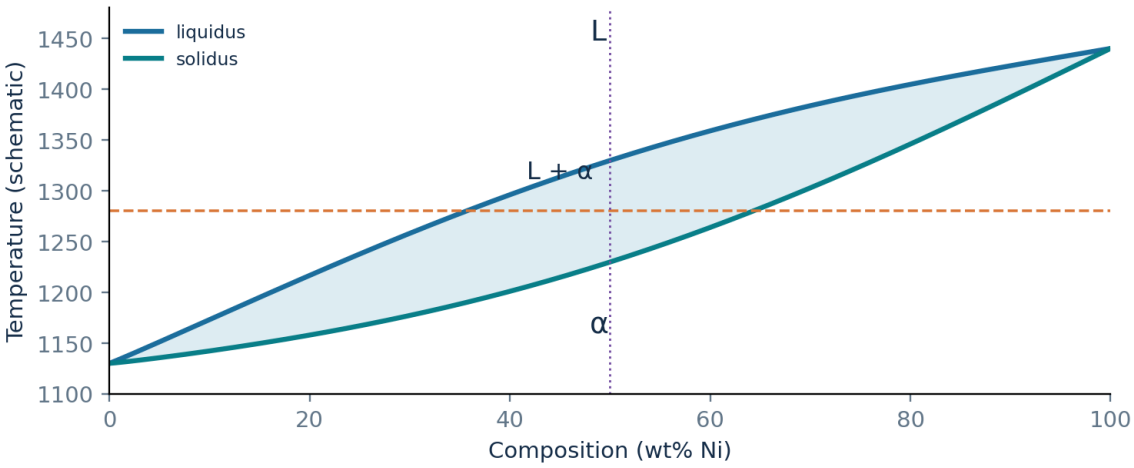
At a specified **overall composition C_0** and **temperature T**, a phase diagram predicts which equilibrium phases exist and, in a two-phase region, their compositions and relative amounts. It is a map of **equilibrium states**. Real cooling may be too fast to reach equilibrium, leading to composition gradients and different microstructures.

The lecturer introduces **Hume–Rothery guidelines** for extensive substitutional solid solubility: atomic radii differ by less than ~15%, crystal structure is the same, electronegativities are similar, and valences are similar/the same. The Cu–Ni pair satisfies these well and is mutually soluble over the whole composition range in solid and liquid states.

> **HUME–ROTHERY MEMORY**
> **Size | Structure | Electronegativity | Valence.** The rules are useful guidelines for extensive solubility, not a complete thermodynamic guarantee for every material system.

Isomorphous system: only one solid-solution phase

Cu–Ni is an **isomorphous** binary system. Its phase diagram contains a liquid field L, an FCC solid-solution field α, and a two-phase field α + L in between. Cu–Ni alloys can remain a **single FCC solid solution** at all compositions once fully solidified at the lower temperatures discussed in the lecture.



Generic isomorphous temperature-composition diagram (not measured Cu–Ni data). Horizontal dashed line illustrates how to read phase compositions from liquidus and solidus.

Three separate questions on the same graph

- **Which phases?** Locate C_0 and T and read the name of the region; do not calculate fractions yet.
- **What composition is each phase?** Draw a horizontal tie line to each boundary and read C_L, C_α. The *overall* C_0 is not generally either phase composition.
- **How much of each phase?** Use the lever rule, taking the opposite tie-line segment and dividing by the entire tie-line length.

$$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L} \qquad W_\alpha = \frac{C_0 - C_L}{C_\alpha - C_L}$$

Symbols: W_L, W_α = mass fractions of liquid and solid α; C_L = liquid composition; C_α = solid composition; C_0 = bulk alloy composition (all on same composition scale).

Condition: Written for $C_L < C_0 < C_\alpha$. If endpoints are reversed, use the same opposite-lever-arm idea or positive interval lengths.

The denominator is the **total tie-line span**, and each numerator is the distance from **C_0 to the opposite phase endpoint**. Check $W_L + W_\alpha = 1$. Convert to percentages only after finding fractions.

METHOD | Full phase-diagram question procedure

01. Mark the x-axis basis (wt% A or wt% B) and temperature axis; verify you have the correct diagram.

02. Mark bulk alloy composition C_0 as a vertical line and the specified T as a horizontal line.

03. Identify the **phase field** at their intersection.

04. If two phases: draw an isothermal tie line and read each phase composition at the boundaries.

05. Find the **opposite** tie-line arm for each phase to calculate its mass fraction.

06. Check fractions 0–1, total = 1 and material balance $C_0 = W_L C_L + W_\alpha C_\alpha$.

07. If asked about cooling, repeat the phase-identification step at successive temperature zones as the vertical C_0 line descends.

WORKED EXAMPLE | Isomorphous lever rule

Question. At a temperature in α + L, a hypothetical alloy has C_L = 30 wt% Ni, C_α = 50 wt% Ni and C_0 = 42 wt% Ni. Find mass fractions.

Step 1. The two phases are liquid and α, with overall C_0 between phase endpoint compositions.

Step 2. Liquid fraction $W_L = (50 - 42)/(50 - 30) = 8/20 = 0.40$.

Step 3. Solid fraction $W_\alpha = (42 - 30)/(50 - 30) = 12/20 = 0.60$.

Step 4. Check $0.40 + 0.60 = 1$ and $(0.40 \times 30) + (0.60 \times 50) = 42$.

> **RESULT**
> **40 wt% liquid** and **60 wt% α**; the two phase compositions remain 30 and 50 wt% Ni.

Equilibrium cooling versus non-equilibrium cooling

On slow **equilibrium cooling**, a liquid Cu–Ni alloy first reaches the liquidus, begins to form α crystallites, and passes through an L + α interval. Tie-line endpoints shift with falling temperature and diffusion allows compositions to adjust. Below the solidus the alloy is fully solid α. Under **non-equilibrium cooling**, limited solid-state diffusion can leave composition gradients within grains (*coring*) and a departure from the equilibrium structure.

METHOD | Describe microstructure along a vertical cooling path

01. Begin **above the liquidus**: homogeneous liquid L of overall composition C_0.

02. At the **liquidus**: first crystals of α nucleate.

03. In **L + α**: the amounts of liquid and α change; each phase composition changes along its boundary as T falls.

04. At the **solidus**: the last liquid solidifies.

05. Below solidus: fully α polycrystalline solid. For rapid cooling mention possible coring/segregation, not a new phase unless a specific diagram requires it.

> **APPLICATION & MEMORY**
> The Cu–Ni diagram can be connected to **solid-solution strengthening**: dissolved Ni in Cu (and Cu in Ni) makes dislocation glide harder, often increasing hardness and strength at intermediate compositions. The slides show strength reaching a maximum somewhere between the pure components while ductility trends in the opposite direction.

A phase is not the same thing as a grain

Many grains of α with different orientations are still only **one phase**. A two-phase microstructure may consist of α and β, each present in many grains or lamellae. Equally, phase **composition** is not the same as phase **amount**. Write clearly which the question requests.

09

Eutectic phase diagrams and microstructure evolution

The final lecture: know what appears first and what transforms at the eutectic point • Lecture basis: Phase Diagrams – Eutectic Systems (15 slides)

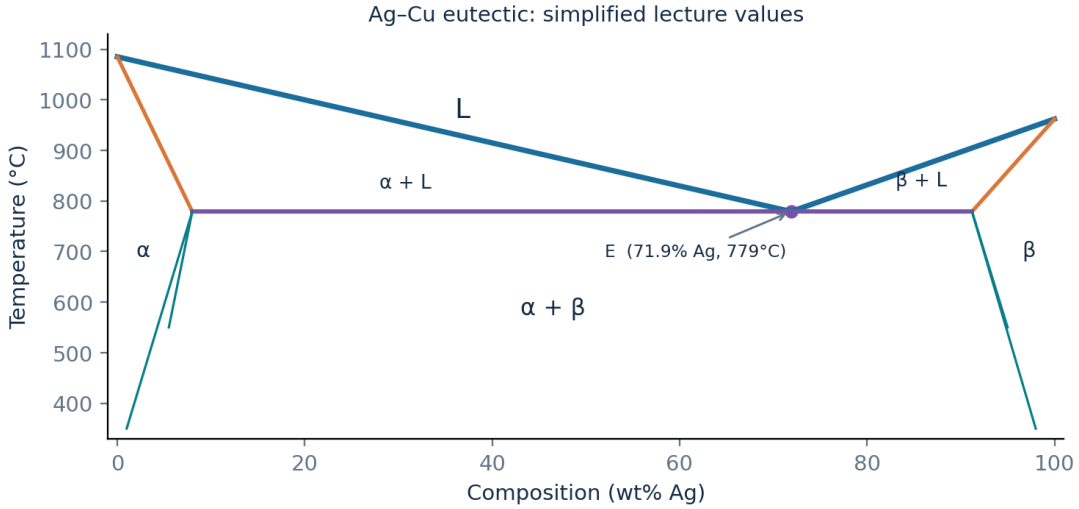
Isomorphous versus eutectic: two different structures

In the Cu–Ni isomorphous system the two components can form **one solid-solution phase α at any proportion**. A **binary eutectic system** has limited mutual solid solubility and normally contains **two distinct solid solutions α and β** plus liquid. The two descending liquidus branches meet at the **eutectic point**, the minimum temperature at which liquid can exist in this kind of simple eutectic system.

Feature	Cu–Ni isomorphous	Ag–Cu / Pb–Sn eutectic
Single solid phases	α only	α rich in left-hand component; β rich in right-hand component
Liquid / solid fields	L; L + α; α	L; L + α; L + β; α; β; α + β
Reaction on cooling	L → L + α → α across a freezing range	At exact eutectic C_E: L → α + β at T_E
Typical final microstructure	One solid-solution phase, many grains	Often primary solid + eutectic α/β mixture, or entirely eutectic mixture

Ag–Cu: memorise the eutectic point and phase definitions

The Ag–Cu lecture uses an x-axis of **weight percent Ag**. α is the Cu-rich FCC solid solution (Ag solute in Cu solvent), while β is the Ag-rich FCC solid solution (Cu solute in Ag solvent). The solid solubility of either metal in the other is limited.



Copper–silver eutectic diagram redrawn from the important lecture values. Boundaries are schematic except the labeled eutectic compositions/temperature.

Ag–Cu anchor point	Value / meaning
Eutectic composition C_E	**71.9 wt% Ag**
Eutectic temperature T_E	**779°C**
α at T_E: $C_{\alpha E}$	**8.0 wt% Ag** (maximum Ag in Cu at eutectic T)
β at T_E: $C_{\beta E}$	**91.2 wt% Ag** (maximum Cu in Ag at eutectic T)
Left and right solid phases	α = Cu rich, β = Ag rich; both FCC
Regions with two phases	**α + L, β + L, α + β**
Boundary names	Liquidus: L/solid + L; solidus: all-solid/solid + L; solvus: single solid/two solids

Why the horizontal eutectic line matters: at T_E the liquid of composition C_E undergoes an isothermal transformation into two solids with the fixed endpoint compositions $C_{\alpha E}$ and $C_{\beta E}$. Liquid cannot remain as an equilibrium phase below the eutectic temperature for alloys within the simple diagram's solid region.

$$L(C_E) \longrightarrow \alpha(C_{\alpha E}) + \beta(C_{\beta E})$$

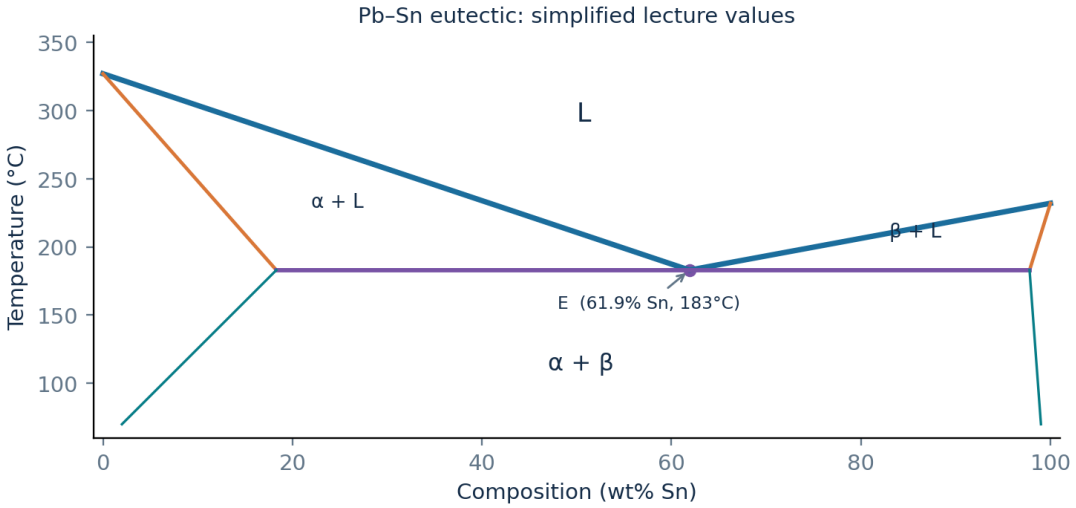
Symbols: L = liquid; α = Cu-rich solid solution; β = Ag-rich solid solution; C_E = eutectic liquid composition; $C_{\alpha E}$, $C_{\beta E}$ = compositions of the two solid phases at eutectic temperature; cooling direction →.

Condition: On heating, the reverse transformation occurs. This is a eutectic reaction, not a eutectoid reaction.

> **LINE LABELS TO RECOGNISE**
> **Liquidus:** boundary separating L from L + solid. **Solidus:** line below which no liquid exists (the eutectic isotherm is part of this boundary). **Solvus:** lower-temperature solid-state solubility boundary. **Eutectic horizontal:** reaction at fixed T_E.

Pb–Sn: memorize its different numerical anchors

The final slides illustrate four different cooling paths on **lead–tin**, a second eutectic system. The composition axis is **weight percent Sn**, so α is Pb-rich and β is Sn-rich.

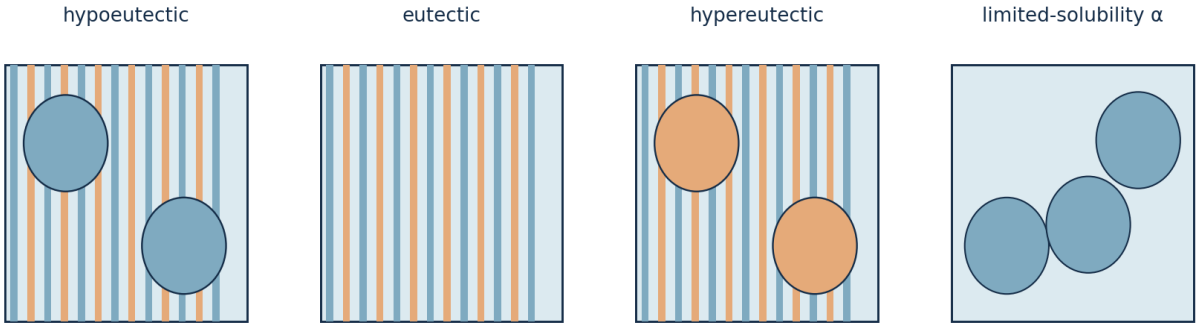


Lead–tin eutectic phase diagram. Notice that the composition axis is wt% Sn, not wt% Ag as on the Ag–Cu figure.

Pb–Sn anchor point	Value / meaning
C_E, T_E	**61.9 wt% Sn** and **183°C**
$C_{\alpha E}$	**18.3 wt% Sn** (Pb-rich α solubility limit at 183°C)
$C_{\beta E}$	**97.8 wt% Sn** (Sn-rich β at 183°C)
Pure component melting points	Pb ≈ **327°C**; Sn ≈ **232°C**

Microstructure words: phases versus microconstituents

The **eutectic microconstituent** is the fine, often lamellar or alternating α + β structure that forms when eutectic liquid transforms at T_E. It contains **two phases**, but is sometimes counted as **one microstructural constituent** because it formed together from that liquid. Meanwhile, **primary / proeutectic α** is the α that forms earlier, above the eutectic temperature. It is chemically still the α phase, even though its shape and formation history differ from α inside the eutectic.



Schematic final microstructures: large primary regions and fine eutectic mixture. Real microstructures depend on processing and cooling rates.

Four cooling paths in the final Pb–Sn lecture

Starting alloy	Cooling sequence and final structure
C1: very Pb-rich, within the α side	L → L + α → α; if later cooling crosses the solvus, the α becomes supersaturated and **β precipitates** in equilibrium. Depending on composition, there may be no eutectic constituent.
C2: Pb-rich, below 18.3 wt% Sn	L → L + α → α before reaching 183°C; after crossing the solvus at a lower temperature, fine β may appear. This is **not automatically** a primary α + eutectic mixture.
C3: 61.9 wt% Sn	Entirely L above 183°C; at 183°C all remaining L transforms into **fine α + β eutectic**. No primary α or β is formed on an ideal equilibrium path.
C4: 18.3 < C_0 < 61.9 wt% Sn	L → L + **primary α**; remaining L approaches C_E, then transforms at 183°C into α + β eutectic. Final: **primary α + eutectic mixture**.

For completeness, an alloy with **61.9 < C_0 < 97.8 wt% Sn** is **hypereutectic**: primary β forms above 183°C, then the remaining liquid transforms into α + β eutectic at the eutectic isotherm. This follows from the same diagram and logic, although the last four slides emphasize the four labeled trajectories above.

METHOD | Describe a eutectic alloy cooling path

01. Identify the system and composition basis: Ag–Cu uses wt% Ag; Pb–Sn uses wt% Sn.

02. Plot a vertical line at the bulk composition C_0; start at a high T in the liquid field.

03. Read each new phase field crossed as T decreases. Name the **primary** solid that appears above the eutectic line, if any.

04. At T_E, ask whether liquid remains. If so, that remaining liquid transforms by $L(C_E) \rightarrow \alpha + \beta$.

05. Below T_E, mention both phase compositions moving along solvus lines if the diagram shows further solid-state change.

06. Describe the final microstructure using precise terms: “α phase,” “β phase,” “primary α/β,” and “eutectic α + β constituent.”

Lever rule for eutectic systems: what exactly is being counted?

Inside any **two-phase field**, use the ordinary lever rule with the two endpoint compositions at the chosen T. However, if the question asks for **primary solid versus eutectic microconstituent**, apply the lever rule at a temperature **just above T_E** to find solid already present versus remaining liquid. The remaining liquid becomes the eutectic constituent on cooling through T_E.

$$W_{\alpha}^{\text{primary}} = \frac{C_E - C_0}{C_E - C_{\alpha E}}$$

Symbols: For a hypoeutectic alloy $C_{\alpha E} < C_0 < C_E$: $W_{\alpha}^{\text{primary}}$ = pre-eutectic α mass fraction; C_0 bulk composition; C_E eutectic liquid composition; $C_{\alpha E}$ α endpoint at T_E.

Condition: Evaluated just ABOVE the eutectic temperature; solid and liquid coexist.

$$W_{\text{eutectic}} = \frac{C_0 - C_{\alpha E}}{C_E - C_{\alpha E}}$$

Symbols: W_{eutectic} = mass fraction of remaining liquid just above T_E, which solidifies as eutectic constituent; symbols as above.

Condition: Primary fraction plus eutectic microconstituent fraction = 1.

$$W_{\alpha} = \frac{C_{\beta E} - C_0}{C_{\beta E} - C_{\alpha E}} \qquad W_{\beta} = \frac{C_0 - C_{\alpha E}}{C_{\beta E} - C_{\alpha E}}$$

Symbols: W_{α}, W_{β} = total phase fractions of α and β just BELOW T_E; $C_{\alpha E}$, $C_{\beta E}$ = equilibrium phase endpoint compositions; C_0 = overall composition.

Condition: The total α includes primary α AND α within the eutectic; microconstituent fractions do not equal phase fractions.

WORKED EXAMPLE | Pb–Sn: primary α and eutectic amount

Question. A Pb–Sn alloy contains C_0 = 40 wt% Sn. Using the lecture values at 183°C, find fractions of primary α and eutectic microconstituent just below T_E.

Step 1. This is hypoeutectic because 18.3 < 40 < 61.9 wt% Sn.

Step 2. Just above 183°C the phases are primary α (18.3%) and L (61.9%).

Step 3. $W_{\text{primary }\alpha}$ = (61.9 − 40)/(61.9 − 18.3) = 21.9/43.6 ≈ 0.502.

Step 4. Remaining liquid (future eutectic) W_{eutectic} = (40 − 18.3)/43.6 ≈ 0.498.

Step 5. As this liquid transforms, the final microstructure contains approximately 50.2% primary α plus 49.8% eutectic constituent by mass.

> **RESULT**
> **50.2% primary α** and **49.8% eutectic microconstituent**. This is not the same as total α versus β phase fractions.

WORKED EXAMPLE | Pb–Sn: total α and β at eutectic temperature

Question. For the same 40 wt% Sn alloy, find total α and β mass fractions just below 183°C.

Step 1. Now use the two SOLID phase endpoints: $C_{\alpha E}$ = 18.3 and $C_{\beta E}$ = 97.8 wt% Sn.

Step 2. $W_{\alpha} = (97.8 - 40)/(97.8 - 18.3) = 57.8/79.5 \approx 0.727$.

Step 3. $W_{\beta} = (40 - 18.3)/79.5 = 21.7/79.5 \approx 0.273$.

Step 4. The α amount includes both the coarse primary α and the fine α that forms within the eutectic mixture.

RESULT
Total α ≈ 72.7%, total β ≈ 27.3%. Compare these with 50.2% primary α and 49.8% eutectic microconstituent.

How to calculate α and β inside the eutectic constituent

$$W_{\alpha|e} = \frac{C_{\beta E} - C_E}{C_{\beta E} - C_{\alpha E}} \qquad W_{\beta|e} = \frac{C_E - C_{\alpha E}}{C_{\beta E} - C_{\alpha E}}$$

Symbols: $W_{\alpha|e}$, $W_{\beta|e}$ = phase mass fractions INSIDE a eutectic microconstituent only; C_E, $C_{\alpha E}$, $C_{\beta E}$ = the three eutectic anchor compositions.

Condition: Use this only when the question asks about the internal phases of eutectic, not the whole hypoeutectic alloy.

WORKED EXAMPLE | Phases within a Pb–Sn eutectic colony

Question. At 183°C, a eutectic colony has overall C_E = 61.9 wt% Sn, with α at 18.3 and β at 97.8. Find the α fraction within that colony.

Step 1. Within the eutectic constituent, $W_{\alpha|e} = (97.8 - 61.9)/(97.8 - 18.3)$.

Step 2. $W_{\alpha|e} = 35.9/79.5 \approx 0.452$.

Step 3. $W_{\beta|e} = 1 - 0.452 = 0.548$.

RESULT
Each eutectic microconstituent contains about **45.2% α** and **54.8% β** by mass at the eutectic temperature.

THE BIGGEST EUTECTIC EXAM TRAP
Phase fractions: total α and β masses. **Microconstituent fractions:** amount of primary solid and amount of eutectic mixture. The eutectic mixture itself contains two phases. Always read the wording before choosing a lever rule.

Cooling-path comparison in one glance

Bulk composition relative to eutectic	First solid on cooling	Final microstructure near/below T_E
Exactly C_E	None before T_E	Eutectic α + β only
$C_{\alpha E} < C_0 < C_E$	Primary α	Primary α + eutectic α + β
$C_E < C_0 < C_{\beta E}$	Primary β	Primary β + eutectic α + β
Very near pure left or right	Respective terminal solid solution	Single solid solution and possibly later precipitates after crossing solvus

10

The exam room toolkit

A practical final review: identify the question, pick the method, check the answer • Lecture basis: All nine lectures

Use the question itself to choose the tool

Question trigger	First thought and method
“Bonding, melting point, expansion”	Compare bond energy well, stiffness (slope) and primary/secondary bond types.
“Atoms per unit cell, APF, atomic radius”	Name crystal structure, count shares, find touching diagonal, then APF.
“Direction, plane, family”	Directions = head − tail → [ ]; planes = intercept reciprocal → ( ); families use ⟨ ⟩ or { }.
“Densest direction/plane”	Atom centers on line/plane divided by segment length/plane area.
“Vacancies at a temperature”	Convert to K, match eV/J units, obtain site density, exponential law.
“Why ductile? why strengthen?”	Dislocation glide; explain barriers from grain boundaries, solute or cold work.
“Yield, modulus, ductility, toughness”	Read the stress–strain curve; slope, offset, max stress, area and fracture dimensions.
“Percent cold work”	Original area vs deformed area; not tensile fracture area.
“Name phases at C_0 and T”	Plot point in phase diagram; report phase field.
“Composition of phases”	Draw horizontal tie line and read both endpoints.
“Fraction of phases”	Lever rule = opposite arm / total arm.
“Cooling / microstructure”	Follow vertical composition path downward; explain first solid and eutectic reaction.
“Primary and eutectic amounts”	Take lever rule JUST ABOVE eutectic temperature.

Question trigger	First thought and method
“Total α and β after eutectic”	Use BOTH solid compositions JUST BELOW eutectic temperature.

Mistakes you should actively avoid

- **MPa conversion:** 1 MPa = 1 N/mm², so area in mm² and force in N lead directly to MPa.
- **Kelvin:** vacancy formula requires T in kelvin, never directly in °C.
- **Exponent units:** energy Q_v and k must both use compatible joules or electronvolts.
- **FCC/BCC diagonal:** FCC face diagonal √2a = 4r; BCC body diagonal √3a = 4r.
- **Atom centers only:** no atoms are counted in density unless their center lies on the selected line or plane.
- **Plane indices:** reciprocate axis intercepts; direction indices are not reciprocals.
- **Yield strain:** 0.2% = 0.002 in decimal strain—not 0.2.
- **True versus engineering stress:** only true stress uses instantaneous area.
- **Phase composition versus overall C_0:** end-of-tie-line values are separate from bulk composition.
- **Lever rule:** opposite-arm length gives the named phase fraction; sum must be 1.
- **Phase versus eutectic constituent:** eutectic is α + β, not a third single phase.
- **What diagram?** use wt% Ag for Ag–Cu and wt% Sn for Pb–Sn; anchor numbers differ.
- **Strengthening versus toughness:** increasing strength frequently reduces ductility; a hard material is not automatically tough.

Practice: attempt these WITHOUT looking at the answers

1. Identify four major classes of engineering materials and one distinct property of each.

2. What force condition holds at an equilibrium interatomic separation?

3. Why do many ionic ceramics have high melting temperatures?

4. Compare a covalent bond with a metallic bond.

5. Give atom count, coordination number and APF of BCC and FCC.

6. A BCC atom has r = 0.125 nm. Estimate a.

7. Find the [uvw] direction from tail (1,0,0) to head (0,1,1).

8. Find (hkl) for intercepts (a,∞,c/2).

9. What is the FCC closest-packed plane and direction?

10. Does raising temperature raise or lower equilibrium vacancies? Why?

11. Compare edge and screw dislocations by b orientation.

12. A tensile test uses F = 12 kN, A_0 = 60 mm². What is stress?

13. How is 0.2% offset yield determined on a curve?

14. Why can the engineering stress drop after UTS while true stress rises?

15. A wire area drops 100 → 75 mm². Calculate percent cold work.

16. Why does finer grain size often strengthen a metal?

17. List Hume–Rothery guidelines.

18. For C_L = 20, C_0 = 35, C_α = 50 wt%, what is W_L?

19. State Ag–Cu eutectic composition, temperature and solid compositions.

20. Explain the final equilibrium microstructure of Pb–Sn at 61.9 wt% Sn.

21. In Pb–Sn, what is the first solid for 40 wt% Sn when cooling from liquid?

22. What is the distinction between total α fraction and primary α fraction?

Answer key: check your reasoning, not only the number

1. Metals: conductive/ductile; ceramics: heat-resistant/hard but brittle; polymers: low density/easily formed; composites: properties tailored by combined constituents.

2. At $r = r_0$, net interatomic force $F_N = F_A + F_R = 0$; potential energy is minimum.

3. Strong electrostatic attraction/large bond-separation energy generally requires substantial thermal energy to separate ions.

4. Covalent: localized directional electron sharing; metallic: nondirectional ion cores bonded by delocalized mobile electron cloud.

5. BCC: n = 2, CN = 8, APF = 0.680; FCC: n = 4, CN = 12, APF = 0.740.

6. $a = 4r/\sqrt{3} = 4(0.125)/1.732 \approx 0.2887$ nm.

7. Head − tail = (−1,1,1) → [−1 1 1].

8. Normalize (1,∞,1/2); reciprocate (1,0,2) → (102).

9. {111} plane family and ⟨110⟩ direction family.

10. It raises $N_v = N \exp[-Q_v/(kT)]$; exponent becomes less negative as T increases.

11. Edge: Burgers vector perpendicular to line; screw: Burgers vector parallel to line.

12. $\sigma = 12{,}000/60 = 200$ N/mm² = 200 MPa.

13. Mark ε = 0.002, draw line parallel to linear elastic slope, read intersection stress.

14. A neck makes actual cross-section smaller; engineering stress still divides by original A_0.

15. [(100 − 75)/100] × 100 = 25%.

16. More grain boundaries obstruct dislocation motion; Hall–Petch $\sigma_y = \sigma_0 + k/\sqrt{d}$.

17. Atomic radius mismatch roughly less than 15%; same crystal structure; similar electronegativity; similar/same valence.

18. $W_L = (50 - 35)/(50 - 20) = 0.50$; $W_\alpha = 0.50$.

19. C_E = 71.9 wt% Ag, T_E = 779°C; $C_{\alpha E}$ = 8.0 wt% Ag, $C_{\beta E}$ = 91.2 wt% Ag.

20. At 183°C, all liquid turns into eutectic α + β; no primary α or β before reaction.

21. Primary Pb-rich α; remaining liquid later makes eutectic α + β at 183°C.

22. Total α includes primary α plus the α inside eutectic; primary α is only solid formed before the eutectic reaction.

The day-before-exam checklist

☐ **01** Explain bonding mechanisms, related properties, and bonding-energy curves without memorizing only a slogan.

☐ **02** Draw SC, BCC and FCC cells; prove their a/r relationships and APFs; remember HCP stacking.

☐ **03** Calculate crystallographic directions and Miller indices; explain linear and planar atomic densities.

☐ **04** Describe every major crystal defect and how a dislocation produces slip.

☐ **05** Sketch and label a tensile stress–strain curve and calculate every property from it.

☐ **06** Explain the physical mechanism and equation for each strengthening method.

☐ **07** Recognize Cu–Ni phase regions; determine phase compositions and lever-rule mass fractions.

☐ **08** Draw/interpret Ag–Cu and Pb–Sn eutectics; remember both systems' distinct eutectic anchor values.

☐ **09** Explain equilibrium cooling paths and tell microconstituents apart from phases.

☐ **10** Repeat the worked calculations without the steps; verify units, fractions and significant figures.

> **FINAL EXAM STRATEGY**
> Study the underlying mechanism first. In a numerical answer, write **Given → Equation → Substitution → Units → Interpretation**. In a theory answer, write **Definition → Mechanism → Consequence → Example**. For a phase-diagram question, state **phases → compositions → fractions → cooling changes** in that order.

Lecture-page traceability (coverage map)

Uploaded lecture	Where it appears in this guide
Introduction I (10 p.) and II (19 p.)	Chapter 1, shortened as requested.
Atomic Bonding (31 p.)	Chapter 2: force, energy, 3 primary + 4 secondary categories, mixed ionic character.
Crystal Structures (66 p.)	Chapters 3 and 4: all systems, SC/BCC/FCC/HCP, directions, planes, linear/planar density.
Crystal Defects (30 p.)	Chapter 5: four point defects, vacancies, dislocations, slip, surface defects.
Mechanical Properties (37 p.)	Chapter 6: full tensile graph, true stress, ductility, brittle testing, Charpy/Izod, hardness.
Strengthening (28 p.)	Chapter 7: grain size, Hall–Petch limits, substitutional solutes, cold working.
Phase Diagrams (37 p.)	Chapter 8: terminology, Hume–Rothery, Cu–Ni, tie line, lever rule and cooling.
Eutectic Systems (15 p.)	Chapter 9: Ag–Cu/Pb–Sn, reaction, four cooling types, microconstituent and phase fractions.

Source note: Diagrams are explanatory redrawings; some numerical exercises use hypothetical values, identified in their questions. Supporting mathematical consequences are explained in context.

LAST 10 MINUTES | MT1011 MEMORY BOARD

Use this only **after** learning the reasoning. If something here feels unfamiliar, return to its chapter.

Cue	Immediate recall
Bonding	Ionic: transfer, nondirectional. Covalent: sharing, directional. Metallic: mobile electron sea, nondirectional. Secondary: dipole attraction.
Unit cells	SC n=1/CN6/APF .52/a=2r; BCC n=2/CN8/APF .68/√3a=4r; FCC n=4/CN12/APF .74/√2a=4r; HCP n=6/CN12/APF .74.
Indices	[uvw] = head minus tail, clear fractions. (hkl) = reciprocal intercepts. [ ] vector; ( ) plane; ⟨ ⟩ and { } family.
Atomic density	ρL = effective centers / segment length; ρP = effective centers / plane area. FCC close-packed: ⟨110⟩, {111}.
Vacancies	Nv = N exp[−Qv/(kT)]; T in K, energy units consistent, site count N = ρNA/M.
Stress–strain	σe = F/A₀; εe = Δl/l₀; E = slope; 0.2% offset = 0.002 strain; UTS = max engineering stress.
Ductility	%EL = (lf−l₀)/l₀ ×100; %RA = (A₀−Af)/A₀ ×100; after UTS necking begins.
Strengthening	Finer grains → more boundaries (Hall–Petch). Solutes → lattice strains. Cold work → higher dislocation density; %CW = (A₀−Ad)/A₀ ×100.
Lever rule	At given T, read tie-line endpoints first. Phase fraction = opposite tie-line arm / total arm. Fractions sum 1.
Ag–Cu eutectic	779°C, 71.9 wt% Ag; α at 8.0 wt% Ag; β at 91.2 wt% Ag.
Pb–Sn eutectic	183°C, 61.9 wt% Sn; α at 18.3 wt% Sn; β at 97.8 wt% Sn.
Cooling path	C₀ < CE: primary α (if above α solubility). C₀ = CE: only eutectic α+β. C₀ > CE: primary β (if below β solubility).
Microconstituents	Primary α fraction ≠ total α fraction. Remaining liquid just above TE becomes eutectic mixture (which contains BOTH α and β).

A GOOD ANSWER IS A CHAIN OF REASONS

Theory: define → explain mechanism → connect to property → give a relevant example. **Calculation:** identify model → write symbols and units → substitute → verify physically. **Phase diagram:** phases → tie-line compositions → phase fractions → cooling microstructure.